



NEW L. J. INSTITUTE OF ENGINEERING AND TECHNOLOGY

SEMESTER: I

**DEPARTMENT
OF
COMPUTER SCIENCE AND ENGINEERING**

QUESTION BANK

SUBJECT CODE: BE01R00121

**SUBJECT NAME: PROGRAMMING FOR PROBLEM
SOLVING**

Exploring Emerging Technologies

Subject Code:	BE01R00121	Subject Name	Credit:	4
Semester:	I	Programming for Problem Solving	Hours:	120
Department:	Computer Science and Engineering			

CHAPTER: 1**INTRODUCTION TO PROGRAMMING**

Introduction to Programming • Basics of Computers: Overview of computer systems, hardware, and software. • Introduction to Programming Languages: Types of programming languages, language translators (compilers, interpreters). • Flowcharts and Algorithms: Basics of creating flowcharts and writing algorithms for problem-solving.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	List out and briefly explain basic building blocks of Computer system.(Mar-2023)[NLJIET]	3	CO2	RM
2	Design block diagram of computer and explain functionalities of various components of computer.(Aug-2023)[NLJIET]	3	CO2	CR
3	Recall the definitions of (I) Assembler (II) Compiler (III) Interpreter. [NLJIET]	3	CO2	RM
4	What is Software and Hardware? Explain different types of Software. [NLJIET]	3	CO2	UN
5	What is Software? List various types of Software. (Jan-2025) [NLJIET]	3	CO2	UN
6	List out types of software with examples. (Jan-2020) [NLJIET]	3	CO2	RM
7	What is the purpose of compiler? Differentiate compiler and interpreter.[NLJIET]	3	CO2	UN
8	Define (1) Higher Level Language (2) Lower Level Language (3) Machine Level Language. [NLJIET]	3	CO2	RM
9	Draw the flow chart to find the largest of the given three numbers – A ,B and C (Mar-2022) [NLJIET]	3	CO2	CR
10	Define algorithm and explain different symbols used in flowchart. [NLJIET]	3	CO2	RM
11	Write an algorithm for finding odd and even number from given two numbers (Mar-2022) [NLJIET]	3	CO2	CR
12	Explain flow chart with suitable example. [NLJIET]	3	CO2	RM
13	Define: algorithm, flowchart, and compiler(Jan-2019)[NLJIET]	3	CO2	RM
14	Explain different symbols used in flowchart.(Nov-2020) [NLJIET]	3	CO2	RM
15	Write an algorithm to determine whether the given year is a leap year or not. [NLJIET]	3	CO2	CR
16	Write an algorithm to check whether the entered number is Even or Odd.(Jan-2024) (Jan-2025-New)[NLJIET]	3	CO2	CR
17	Write flowchart or algorithm to find area of a triangle. (Jun-2019) [NLJIET]	3	CO2	CR
18	Draw neat Block Diagram illustrating the Anatomy of a	3	CO2	CR

	Computer System.(Jul-2024)[NLJIET]			
19	Draw block diagram of computer and explain functionalities of various components of computer. (Jun-2025-New) [NLJIET]	3	CO1	UN
20	Brief about computer hardware and software. (Jun-2025-New) [NLJIET]	4	CO1	UN
21	Define following terms: 1. Compiler 2. Runtime Error 3. Dynamic Memory Allocation 4. Recursion (Jan-2025) [NLJIET]	4	CO2	RM
22	Categorize the major components of computer system and give their function. (Jan-2020) [NLJIET]	4	CO2	UN
23	Write algorithm or flowchart for finding maximum out of three numbers entered by user. (Nov-2020) [NLJIET]	4	CO2	CR
24	Write an algorithm to find maximum number for three given number. (Aug-2023)[NLJIET]	4	CO2	CR
25	Draw flowchart to find out maximum number out of three numbers.[NLJIET]	4	CO2	CR
26	Write an algorithm and draw a flowchart to print first N Fibonacci numbers. [NLJIET]	4	CO2	CR
27	Develop an algorithm to print first N Fibonacci numbers. (Jan-2020) [NLJIET]	4	CO2	CR
28	Define algorithm and explain different symbols used in flowchart. (Jan-2020)[NLJIET]	4	CO2	UN
29	Describe various symbols used for preparing flow chart.(Mar-2023) (Jan-2025-New)[NLJIET]	4	CO2	RM
30	Explain flow chart with suitable example. [NLJIET]	4	CO2	RM
31	Draw a flow chart that shows how to find root of quadratic equation ax^2+bx+c . [NLJIET]	4	CO2	CR
32	Draw all symbols used in flowchart and draw flowchart to find factorial number.(Jan-2019) [NLJIET]	4	CO2	CR
33	Draw the flowchart to find the factorial of a number given by user (Jan-2024)[NLJIET]	4	CO2	CR
34	Sketch flowchart to convert seconds into hour, minute and second. (Hint: 3700 seconds => 1 hr , 1 minute , 40 second). [NLJIET]	4	CO2	CR
35	What is a flow chart? How is it different from an algorithm? [NLJIET]	4	CO2	UN
36	Illustrate an Algorithm to check whether the given year is leap year or not.(Jul-2024)[NLJIET]	4	CO2	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Draw and explain the block diagram of Computer System. [NLJIET]	7	CO2	UN
2	Draw block diagram of computer system and explain the functions of each component in detail.	7	CO2	UN

	(Aug-2022)[NLJIET]			
3	Briefly explain different components of Computer system (Jan-2024)[NLJIET]	7	CO2	UN
4	Define following: 1) Operating System 2) Compiler 3) Interpreter 4) Assembler 5) Loader 6) Linker 7) Pre-processor. [NLJIET]	7	CO2	RM
5	Write an algorithm and draw the flow chart to find the largest of the given three numbers A ,B and C. [NLJIET]	7	CO2	CR
6	Draw a flow chart to do the sum of 10 elements read from the user. [NLJIET]	7	CO2	CR
7	Write an algorithm and draw a flowchart to print first N Fibonacci numbers. [NLJIET]	7	CO2	CR
8	List all symbols used in flowchart and draw flowchart to find factorial number. (Jan-2020) [NLJIET]	7	CO2	RM
9	Design a flowchart for checking whether a given number is palindrome or not. (Aug-2022)[NLJIET]	7	CO2	CR
10	Define Flow chart, List the rules for designing a Flow chart and Rephrase the merits and demerits of flow chart. (Jul-2024)[NLJIET]	7	CO2	RM
11	Write an algorithm and draw the flowchart to calculate sum of digits of a given number N. (Jan-2025) [NLJIET]	7	CO2	CR
12	Write algorithm or flowchart for finding maximum out of three numbers entered by user. (Jun-2025-New) [NLJIET]	7	CO1	AP

CHAPTER: 2

BASICS OF C PROGRAMMING

Structure of a Program: Basic syntax, semantics, and structure of C/Python programs. • Data Types and Variables: Primitive data types, variables, and constants. • Operators: Arithmetic, relational, logical, bitwise, and assignment operators.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Discuss the important of stdio.h header file. [NLJIET]	3	CO1	AN
2	Show the important of stdio.h header file. (Jan-2020) [NLJIET]	3	CO1	AP
3	What are header files? Name at least 3 with its usage. (Jun-2019) [NLJIET]	3	CO1	RM
4	List any three header file with its usage. (Nov-2020)(Aug-2023)[NLJIET]	3	CO1	RM
5	Explain the different data types of C language. [NLJIET]	3	CO1	UN
6	Explain basic data types of C. [NLJIET]	3	CO1	UN
7	Explain fundamental data types in C language [NLJIET]	3	CO1	UN

8	Explain Derive data types. [NLJIET]	3	CO1	UN
9	Describe the four basic data types. How could we extend the range of values they represent? [NLJIET]	3	CO1	RM
10	List the data types provided by C programming language. [NLJIET]	3	CO1	RM
11	Explain different types of constants. (Jan-2020) [NLJIET]	3	CO1	UN
12	What is formatted output? Using printf() statement explain it. (Jun-2019) [NLJIET]	3	CO1	UN
13	Briefly explain storage class <i>auto</i> and <i>extern</i> . (Mar-2023)[NLJIET]	3	CO1	UN
14	What do you mean by type conversion? Why is it necessary? [NLJIET]	3	CO1	UN
15	What do you mean by type conversion? Give example. (Jan-2019) [NLJIET]	3	CO1	UN
16	What is type conversion? Briefly explain with suitable example. [NLJIET]	3	CO1	UN
17	Explain type casting with example. (Aug-2022)[NLJIET]	3	CO1	UN
18	Explain Type casting in C with example (Aug-2023)[NLJIET]	3	CO1	UN
19	Explain ternary (?:) operator in detail with the example. (Aug-2022)[NLJIET]	3	CO1	UN
20	Demonstrate the use of ternary operator with an example (Jan-2024)[NLJIET]	3	CO1	AP
21	Demonstrate the working of bitwise shift and <i>sizeof</i> operator with an example. (Mar-2023)[NLJIET]	3	CO1	AP
22	List the rules to assign a name to an Identifier (Variable). (Jul-2024)[NLJIET]	3	CO1	RM
23	Explain the basic structure of a C program with an example. (Jan-2025)[NLJIET]	3	CO1	UN
24	Demonstrate the use of conditional operator statement with an example. (Jan-2025-New)[NLJIET]	3	CO1	AP
25	Explain the structure of switch case statement. (Jan-2025-New)[NLJIET]	4	CO1	UN
26	Evaluate given C programming Expressions : 1. Ans = $10 + 4 * 3 / 2$; Explain the structure of switch case statement. 2. Ans = while(1); 3. Ans = $3 * 2 > 3 + 1$; 4. Ans = $(2 + 4) > 3 \&\& 2 < 4$; (Jan-2025)[NLJIET]	4	CO1	AP
27	List the header files used in C Programing with their properties. [NLJIET]	4	CO1	RM
28	Discuss the important of stdio.h and conio.h header file. [NLJIET]	4	CO1	AN
29	Explain following pre-processor directives with example. [NLJIET] 1. #define 2. #ifdef	4	CO1	UN

30	Distinguish the data types provided by C programming language. (Jan-2020)[NLJIET]	4	CO1	AP
31	What is difference between keywords and identifiers? Explain rules for naming an identifier. [NLJIET]	4	CO1	AP
32	Describe the four basic data types. How could we extend the range of values they represent? [NLJIET]	4	CO1	UN
33	Explain four basic data types in C and provision to extend range for appropriate data types. (Nov-2020) [NLJIET]	4	CO1	UN
34	Define variable and constant. Explain different types of constants. [NLJIET]	4	CO1	RM
35	Describe Enumerated data type with example. [NLJIET]	4	CO1	UN
36	List out the operators used in C language and explain any three with example.(Jan-2020) [NLJIET]	4	CO1	RM
37	List out all operators and explain any two in details. (Aug-2023)[NLJIET]	4	CO1	RM
38	Describe precedence and associativity of operators with example. (Jan-2019)[NLJIET]	4	CO1	UN
39	Explain type conversion and type casting with example. (Nov-2020) [NLJIET]	4	CO1	UN
40	What is type conversion? Explain two types of conversion with examples. (Jan-2025)[NLJIET]	4	CO1	UN
41	Demonstrate the use of bitwise operator with an example (Jan-2024)[NLJIET]	4	CO1	AP
42	Briefly explain <<(shift left) and >>(shift right) operator with suitable example. [NLJIET]	4	CO1	UN
43	Give the output of following C code.(Jan-2024)[NLJIET] <pre>int main(){ printf("%d",15<2); printf("%d",15&&2); printf("%d",7%10); return0; }</pre>	4	CO1	EL
44	Give the output of following C code. <pre>int main(){ int val = 20, *p; p=& val printf("%d %d %d", val, * p sizeof(p)); return 0; } (Jan-2024)[NLJIET]</pre>	4	CO1	EL
45	Illustrate the basic structure of a C Programs and highlight usual sections of main function.(Jul-2024)[NLJIET]	4	CO1	AP

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain C tokens in detail(Jan-2024)[NLJIET]	7	CO1	UN
2	Explain primary data types used in C. [NLJIET]	7	CO1	UN

3	List and explain different data types available in C. (Mar-2023)[NLJIET]	7	CO1	UN
4	Briefly explain different storage classes used in C with appropriate example(Jan-2024)[NLJIET]	7	CO1	UN
5	Explain various Operators used in C language. [NLJIET]	7	CO1	UN
6	Discuss about various operators used in C language. [NLJIET]	7	CO1	UN
7	List all operators used in C language and explain any three with example. (Jan-2019) [NLJIET]	7	CO1	RM
8	What is various types of operator available in C. [NLJIET]	7	CO1	RM
9	Explain different categories of operators available in C programming. (Nov-2020)[NLJIET]	7	CO1	UN
10	Explain different type of operators used in c language with their precedence and associativity. (Mar-2022) (Jan-2025-New)[NLJIET]	7	CO1	UN
11	Describe assignment, sizeof() and ternary operator with example [NLJIET]	7	CO1	UN
12	Explain the working of various bit-wise operators with the example.(Aug-2022)[NLJIET]	7	CO1	UN
13	Explain different bit-wise operators with the example. (Jun-2025-New) [NLJIET]	7	CO2	AP

PROGRAMS

1	Write C program to find area and perimeter of rectangle. [NLJIET]	3	CO1	CR
2	Write a program to find area of triangle. (Nov-2020) [NLJIET]	3	CO1	CR
3	Write the outputs of the following expression. (Aug-2022)[NLJIET] i) $50 \% 2 / 3 + 2$ ii) $21 / (\text{int}) 2.5 + 3$ iii) $(1 > 2) \parallel (2 < 3) \&\& 5 < 1$	3	CO1	CR
4	Write a program that reads two numbers from key board and gives their addition, subtraction, multiplication, division and modulo. [NLJIET]	4	CO1	CR
5	Implement a C Program to convert temperature from Fahrenheit to Celsius and vice versa. (Jan-2020)[NLJIET]	7	CO1	CR
6	Write a C program to compute Fahrenheit from centigrade ($f=1.8*c+32$). (Aug-2023)[NLJIET]	7	CO1	CR
7	Write a C program to convert Celsius to Fahrenheit and vice versa. [NLJIET]	7	CO1	CR
8	Write a C program that takes three coefficients (a, b and c) of a quadratic equation ; $(a.x^2 + b.x + c)$ as input and compute all possible roots and print them with appropriate messages. (Jan-2025)[NLJIET]	7	CO1	CR

CHAPTER: 3**CONTROL STRUCTURES**

Conditional Statements: if, if-else, nested if-else, switch-case. • Loops: for, while, do-while loops. •

Break and Continue Statements: Usage in loop control.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain switch case in 'C' with the help of an example. [NLJIET]	3	CO1	UN
2	Explain else...if ladder with an example (Aug-2023)[NLJIET]	3	CO1	UN
3	State the difference between entry control loop and exit control loop. [NLJIET]	3	CO1	AN
4	Explain entry controlled and exit controlled loop with proper syntax. (Nov-2020)[NLJIET]	3	CO1	UN
5	Differentiate: Exit controlled Loop and Entry controlled Loop. (Aug-2023) (Jan-2025-New)[NLJIET]	3	CO1	UN
6	List down three constructs for performing loop operations in C language. Write general form of same. [NLJIET]	3	CO1	RM
7	Compare for loop and while loop with illustrations. [NLJIET]	3	CO1	UN
8	Discuss general form of following decision – making statements. (I) IF (II) SWITCH (III) GOTO. [NLJIET]	3	CO1	UN
9	Define general form of 1) do while loop 2) Nested if 3) goto. (Jan-2019) [NLJIET]	3	CO1	RM
10	Define break and continue statement with example. (Jan-2020) [NLJIET]	3	CO1	RM
11	Discuss use of break and continue statement in C with example. (Mar-2022) [NLJIET]	3	CO1	UN
12	Differentiate between break and continue. (Jan-2024)[NLJIET]	3	CO1	UN
13	Explain following terms: a) break statement b) getch() function c) getchar() function (Aug-2023)[NLJIET]	3	CO1	UN
14	Contrast the Entry Control Loop and Exit Control loop.(Jul-2024)[NLJIET]	3	CO1	UN
15	Differentiate while loop and do..while loop with example. (Jan-2025) [NLJIET]	3	CO1	UN
16	Explain the structure of for loop with proper example. (Jun-2025-New) [NLJIET]	3	CO2	UN
17	Discuss the need of break and continue statements with example. (Jun-2025-New) [NLJIET]	3	CO3	UN

18	Explain while loop and do-while loop with example. (Jun-2025-New) [NLJIET]	3	CO3	UN
19	Explain nested if-else statement with suitable example. (Jun-2025-New)	4	CO2	AP
20	Describe switch case statement with example. [NLJIET]	4	CO1	RM
21	Explain <i>switch</i> statement with an example. (Mar-2023)[NLJIET]	4	CO1	UN
22	Explain nested if else ladder with suitable example. (Mar-2022) [NLJIET]	4	CO1	UN
23	Explain various types of loop available in C with example. [NLJIET]	4	CO1	UN
24	Compare and contrast while and do while loop with example. (Mar-2022) [NLJIET]	4	CO1	UN
25	Explain while loop and do-while loop with example. (Aug-2022)[NLJIET]	4	CO1	UN
26	Differentiate <i>while</i> and <i>do..while</i> loop. (Mar-2023)[NLJIET]	4	CO1	UN
27	Explain entry control loop and exit control loop with example. [NLJIET]	4	CO1	UN
28	Compare entry control loop and exit control loop [NLJIET]	4	CO1	UN
29	What is bottom tested loop? Give example. [NLJIET]	4	CO1	AN
30	Explain for loop with example. (Jan-2020) [NLJIET]	4	CO1	UN
31	Explain the structure of for loop with proper example (Aug-2023)[NLJIET]	4	CO1	UN
32	Explain the use of following statements. (I) if...else (III) goto [NLJIET]	4	CO1	UN
33	Explain: 1. Nesting if-else statement 2. Use of break statement. [NLJIET]	4	CO1	UN
34	Explain 1) nested if-else statement 2) break statement. (Nov-2020) [NLJIET]	4	CO1	UN
35	Explain break and continue statement with example. (Jan-2019)(Jun-2019) [NLJIET]	4	CO1	UN
36	Explain: a) break statement b) continue statement (Jan-2025-New) [NLJIET]	4	CO1	UN
37	Describe break and continue statement with suitable example. [NLJIET]	4	CO1	UN
38	Discuss the need of break and continue statements with example. (Aug-2022)[NLJIET]	4	CO1	AN
39	Compare and contrast goto statement and switch statement with example.(Aug-2022)[NLJIET]	4	CO1	AP
40	Explain the use of switch keyword with proper example. [NLJIET]	4	CO1	UN
41	Explain: a) switch-case statement	4	CO1	UN

	b) continue statement (Aug-2023)[NLJIET]			
42	Demonstrate the use of forward jump and backward jump with an example.(Jan-2024)[NLJIET]	4	CO1	AP
43	Explain nested if-else statement with suitable example.(Jul-2024)[NLJIET]	4	CO1	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain else-if ladder with proper example. [NLJIET]	7	CO1 CO4	UN
2	Explain else... if ladder with an example. (Jan-2024)[NLJIET]	7	CO1 CO4	UN
3	List the various types of loop available in C and explain their syntax.[NLJIET]	7	CO1 CO4	RM
4	Explain various types of loop available in C with example. [NLJIET]	7	CO1 CO4	UN
5	What do you understand by looping? Explain different types of loops in C with example and compare them. [NLJIET]	7	CO1 CO4	UN
6	Explain different loops used in C language. Write the difference between while loop and do-while loop. [NLJIET]	7	CO1 CO4	UN
7	Explain various looping control structures with suitable example.(Jun-2019) [NLJIET]	7	CO1 CO4	UN
8	Explain while loop with example. (Jan-2019) [NLJIET]	7	CO1 CO4	UN
9	Explain for loop with example. (Jan-2019)[NLJIET]	7	CO1 CO4	UN
10	Write a C program to find the largest number amongst three numbers entered by user. (Jun-2025-NEW)[NLJIET]	7	CO1 CO4	CR
11	Write a C program to check if entered integer is a prime number or not. (Jun-2025-NEW)[NLJIET]	7	CO1 CO4	CR

PROGRAMS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Write a program to add first n numbers. Get value of n from user. [NLJIET]	3	CO4	CR

2	Find out error(s), if any in the following code and correct it: (Mar-2023)[NLJIET] int main() { printf(“%f %d”, 7.0%5.0, 5 -2); }	3	CO4	AN
3	Debug the following codes and show the output for this: For (i=0 ; i<=50 ; i++); { printf(“\n %d” , i) } *consider header files and declaration of variable is done. (Jan-2025-New)[NLJIET]	3	CO4	AN
4	Debug the following codes and show the output for this: For (i=0 ; i<=50 ; i++); { printf(“\n %d” , i) } *consider header files and declaration of variable is done. (Jul-2024)[NLJIET]	3	CO4	AN
5	Write a program to select and print the largest of the three nos. using nested-if-else statement. [NLJIET]	4	CO4	CR
6	Write code to find out largest of 2 numbers using ternary operator. [NLJIET]	4	CO4	CR
7	Write a program to print the following pattern. (Aug-2023)[NLJIET] * * * * * * * * * *	4	CO4	CR
8	Write a program to print the triangle shown below. (Aug-2022)[NLJIET] A A B A B C	4	CO4	CR

	A B C D			
9	Write a C Program to check whether the given number is prime or not. [NLJIET]	4	CO4	CR
10	Write a program to reverse a given number. . (Jun-2019) [NLJIET]	4	CO4	CR
11	Write a program to find $1+1/2+1/3+1/4+....+1/n$. . (Jun-2019) [NLJIET]	4	CO4	CR
12	Write a program to print Fibonacci series.e.g.0, 1, 1, 2, 3, 5, 8,13. (Aug-2022)[NLJIET]	4	CO4	CR
13	Examine following code and give output of it. (Mar-2023)[NLJIET] <pre>void main(){ int i=1; for (i=10; i>=1 ; --i) { printf (“ %d”, i); i--; } }</pre>	4	CO4	AN
14	Examine following code and give output of it. (Mar-2023)[NLJIET] <pre>void main(){ int a=0; while (a<10) { a++; if (a%3==1) { continue; } printf (" %d", a); } }</pre>	4	CO4	AN
15	Write a program to print following patterns : (Aug-2023)[NLJIET] 1	4	CO4	CR

	22 333 4444			
16	Write a program which takes 20 numbers from user and display maximum, minimum and average of those numbers.[NLJIET]	4	CO4	CR
17	Give the output of following C codes. (Jan-2024)[NLJIET] <pre> 1. int main(){ int i=10; while(i<10){ printf("%d", i); i++; } return 0; } 2. int main(){ int i=10; do{ printf("%d", i); i++; }while(i<10); return 0; } </pre>	4	CO4	CR
18	Construct a C program to find the largest number amongst three integers entered.(Jul-2024)[NLJIET]	4	CO4	CR
19	Write a program to print the following pattern. [NLJIET] <pre> 1 0 1 1 0 1 0 1 0 1 </pre>	7	CO4	CR

20	Write a program to print the following pattern. [NLJIET] * * * * * * * * * *	7	CO4	CR
21	Write a program to print the following pattern [NLJIET] 1 2 2 3 3 3 4 4 4 4	7	CO4	CR
22	Write a program to print the following pattern. [NLJIET] * * * * * * * * * *	7	CO4	CR
23	Write a program to find the sum of first N odd numbers. [NLJIET]	7	CO4	CR
24	Write a program in 'C' to print the following pattern. [NLJIET] 1 23 456 78910	7	CO4	CR
25	Write a program in 'C' to print the following pattern. [NLJIET] 12345 2345 345 45 5	7	CO4	CR
26	Write a C program to print following pattern. (Mar-2022) [NLJIET] 1 2 3 4 5 6	7	CO4	CR
27	Write a program in 'C' to print the following pattern using loop statement. (Jan-2020) [NLJIET] 1 2 2 3 3 3 4 4 4 4 5 5 5 5 5	7	CO4	CR

28	Write a C program to display given pattern.[NLJIET] 1 12 1 2 3 1 2 3 4 1 2 3 4 5	7	CO4	CR
29	Write a C program to print following pattern. (Mar-2022) [NLJIET] 1 2 2 3 3 3	7	CO4	CR
30	Write a program that reads a value n and generates the following pattern:[NLJIET] 1 1 2 3 1 2 3 4 5 1 2 3 4 5 6 7 1 2 3 4 5 6 7 8 9	7	CO4	CR
31	Write a C program to find $1 + 1/2! + 1/3! + 1/4! + \dots + 1/n!$. (Nov-2020) [NLJIET]	7	CO4	CR
32	Write a C program to check whether a number is prime or not(Aug-2023)[NLJIET]	7	CO4	CR
33	Write a program to find sum of first N odd numbers. Ex. $1+3+5+7+\dots+N$.(Jan-2020)[NLJIET]	7	CO4	CR
34	Write a C program to find factorial of a given number. (Nov-2020) (Mar-2023)[NLJIET]	7	CO4	CR
35	Write a program to print all Armstrong numbers in a given range. Armstrong number is equal to sum of cubes of its individual digits. For example $153 = 1^3 + 5^3 + 3^3$. So, 153 is Armstrong number. .(Jun-2019) [NLJIET]	7	CO4	CR
36	Write a program to print Fibonacci series. (1,1,2,3,5,.....N) [NLJIET]	7	CO4	CR
37	Write a C program to make sum of digits of a given number. (if input is 145, output should be 10).(Mar-2023)[NLJIET]	7	CO4	CR
38	Write a C program to print following pattern: (Mar-2023)[NLJIET] 1 2 2 3 3 3 4 4 4 4	7	CO4	CR
39	Write a C program to check whether the entered character is alphabet,digit, space or any special character. (Mar-2023)[NLJIET]	7	CO4	CR
40	Write a program which takes the amount of bill and the percentage of GST to be charged on the bill amount from user and display total amount payable.[NLJIET]	7	CO4	CR
41	Write a C program to check whether the entered character is alphabet, digit, space or any special character.	7	CO4	CR

	(Mar-2023)[NLJIET]			
42	Write a C program to check whether a three digits number is palindrome or not. (Aug-2023)[NLJIET]	7	CO4	CR
43	Write a C program to print following pattern using loop. (Jan-2024)[NLJIET] 5 4 4 3 3 3 2 2 2 2 1 1 1 1 1	7	CO4	CR
44	Write a C program to display prime number between 1 to 100 [NLJIET]	7	CO4	CR
45	Develop an application program to convert and print distance between two cities in meters, feet, inches & centimeters. The distance between two cities (In KM) is input through key board. [NLJIET]	7	CO4	CR
46	Write a program to select and print the largest of the three nos. using nested-if-else statement. [NLJIET]	7	CO4	CR
47	Write a menu driven C program for simple calculator. Also draw flowchart. [NLJIET]	7	CO4	CR
48	Develop a simple program to Add, subtract and multiply two numbers using switch statement. [NLJIET]	7	CO4	CR
49	Write a program to perform addition, multiplication, subtraction and division with switch statement. [NLJIET]	7	CO4	CR
50	Write a program to accept start number and end number from the user and print all the numbers in the range. [NLJIET]	7	CO4	CR
51	Explain switch case statement with example to read number between 1 to 7 and print relatively day Sunday to Saturday. (Jan-2019) [NLJIET]	7	CO4	CR
52	Write a C program to read numbers 1 to 7 and print relatively day Sunday to Saturday using switch statement. (Nov-2020) [NLJIET]	7	CO4	CR
53	Demonstrate a C program to input an integer number and check last digit of number is even or odd. (Jan-2019) (Jan-2020) [NLJIET]	7	CO4	CR
54	Construct a C program which receives number of days as input and converts them in to Year, Months and Days. (Jul-2024)[NLJIET]	7	CO4	CR
55	Construct a C program to swap the values of two integers with use of only two variables. (Jul-2024) (Jan-2025-New)[NLJIET]	7	CO4	CR
56	Construct a C program to check if entered integer is a prime number or note. (Jul-2024)[NLJIET]	7	CO4	CR
57	Construct a C program to print following pattern 1 2 3 4 5 6	7	CO4	CR

	7 8 9 10 11 12 13 14 15.....(Jul-2024)[NLJIET]			
58	Write a C program which takes marks (between 0 to 100) of a student for N subjects. It should print grade of a student based on Percentage of Student's marks . "A" for $\geq 85\%$, "B" between $\geq 70\%$ and $< 85\%$, "C" between $\geq 50\%$ and $< 70\%$, "PASS" between $\geq 35\%$ and $< 50\%$, "FAIL", if any one subject marks are < 35 . (Jan-2025)[NLJIET]	7	CO4	CR
59	Write a C program to print the following pattern as output: A B C D E F G H I J (Jan-2025)[NLJIET]	7	CO4	CR
60	Write a C program to find factorial of a number. (Jan-2025-New)[NLJIET]	7	CO4	CR
61	Write a program to print the triangle shown below. 1 1 0 1 0 1 1 0 1 0 1 0 1 0 1 (Jan-2025-New)[NLJIET]	7	CO4	CR

CHAPTER: 4

FUNCTIONS AND MODULAR PROGRAMMING

Defining Functions: Syntax, return types, and parameter passing. • Library Functions: Standard library functions and header files (for C). • Recursion: Basic concepts and examples.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is UDF? Describe advantages of UDF. (Jan-2019) [NLJIET]	3	CO1	RM
2	In user defined function, what is actual argument and formal argument? (Jun-2019)[NLJIET]	3	CO1	RM
3	Compare “call by value” & “call by reference” with suitable example. [NLJIET]	3	CO1	AN
4	Explain the difference between call by reference and call by value with proper example. [NLJIET]	3	CO1	UN
5	Differentiate call by value and call by reference. (Aug-2022)[NLJIET]	3	CO1	UN
6	Difference between <i>call by value</i> and <i>call by reference</i> . (Mar-2023)[NLJIET]	3	CO1	UN

7	What is function? How Call by value and Call by reference passes the argument in function. (Jan-2025) [NLJIET]	3	CO1	UN
8	Describe local and global variable with example. (Jan-2019) [NLJIET]	3	CO1	UN
9	Briefly discuss about scope of variable. [NLJIET]	3	CO1	AN
10	What do you understand by recursive function? Explain with small example. [NLJIET]	3	CO1	AN
11	Explain recursive function with proper syntax with small example. [NLJIET]	3	CO1	UN
12	What do you mean by recursive function? What care must be taken while writing a program with recursive function? (Mar-2022) [NLJIET]	3	CO1	AN
13	What care must be taken while writing a program with recursive function? (Jun-2019) [NLJIET]	3	CO1	UN
14	Differentiate function and macro. (Aug-2022)[NLJIET]	3	CO1	UN
15	Define recursion. List the advantages of recursion. (Jan-2019) [NLJIET]	3	CO1	RM
16	List the advantages of recursion. (Jan-2020) [NLJIET]	3	CO1	RM
17	What are the steps in writing a recursive function in a program? (Jan-2025) [NLJIET]	4	CO1	UN
18	What is user-defined function? Explain actual argument and formal-arguments. [NLJIET]	4	CO1	UN
19	Explain function declaration, function definition and function call. [NLJIET]	4	CO1	UN
20	Classification of User defined function's components (elements). (Jan-2019) (Jan- 2020) [NLJIET]	4	CO1	UN
21	List out the categories of functions of C. Explain any one category with example. [NLJIET]	4	CO1	RM
22	Explain Call by value and Call by reference. [NLJIET]	4	CO1	UN
23	Discuss parameter passing technique used in C with example (Jan-2019)[NLJIET]	4	CO1	AN
24	Differentiate pass by value and pass by reference with example. [NLJIET]	4	CO1	UN
25	Differentiate between call by value and call by reference. (Jan-2024)[NLJIET]	4	CO1	UN
26	Explain inline function with example. [NLJIET]	4	CO1	UN
27	Write a function to swap 2 numbers. [NLJIET]	4	CO1	CR
28	Explain problem solving using recursion with any example. (Nov-2020) [NLJIET]	4	CO1	UN
29	Apply quicksort algorithm to sort the following data. Justify the steps. [NLJIET] 42, 29, 74, 11, 65, 58	4	CO1	AP
30	Explain the concept of recursion with an example. (Mar-2023)[NLJIET]	4	CO1	UN

31	Demonstrate the use of recursion with an example. (Jan-2024)[NLJIET]	4	CO1	AP
32	List the categories of User Defined Functions and Discuss any one of these.(Jul-2024)[NLJIET]	4	CO1	RM
33	Explain recursion with example. (Jun-2025-New) [NLJIET]	4	CO3	AP
34	What are header files? List any three with its usage. (Jun-2025-New) [NLJIET]	4	CO3	UN
35	Differentiate call by value and call by reference. (Jun-2025-New) [NLJIET]	4	CO5	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is user defined function? Explain different categories of user defined function.[NLJIET]	7	CO1	UN
2	What is function? Explain the function definition, function prototype and function call with example. [NLJIET]	7	CO1	UN
3	Explain the function definition, function prototype and function call with relative example. (Jan- 2020) [NLJIET]	7	CO1	UN
4	List out the categories of functions of C. Explain any one category with example. [NLJIET]	7	CO1	RM
5	Describe types of function based on arguments and return type. [NLJIET]	7	CO1	UN
6	Explain various categories of functions with suitable example. [NLJIET]	7	CO1	UN
7	Recall and describe different “categories of functions”. [NLJIET]	7	CO1	UN
8	Describe different categories of user-defined functions. (Mar-2022) [NLJIET]	7	CO1	UN
9	Describe categories of User defined function. (Mar-2023)[NLJIET]	7	CO1	UN
10	Explain call by value (pass by value) and call by reference (pass by reference) with examples in brief. (Jan-2020) [NLJIET]	7	CO1	UN
11	What is user defined function? Explain pass by value and pass by reference in functions with example. (Nov-2020) [NLJIET]	7	CO1	UN
12	What is function? Differentiate: Call by value and Call by reference(Aug-2023)[NLJIET]	7	CO1	UN
13	Explain different type of functions used in c language. Write a program using function to find maximum if two numbers. [NLJIET]	7	CO1	UN
14	What is recursive function? Explain with suitable example. (Jun-2019) [NLJIET]	7	CO1	UN
15	What do you mean by recursive function? Write a program in C to find factorial of a number using recursive function. [NLJIET]	7	CO1	CR

PROGRAMS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Write a program that used user defined function Swap() and interchange the value of two variable. [NLJIET]	7	CO4	CR
2	Create a function to swap the values of two variables. (Jan-2019) [NLJIET]	7	CO4	CR
3	Create a function to check number is prime or not. If number is prime then function return value 1 otherwise return 0. (Jan-2019) (Jan-2020) [NLJIET]	7	CO4	CR
4	Write a function in the program to return 1 if number passed is prime otherwise return 0. (Nov-2020)[NLJIET]	7	CO4	CR
5	Write a program to check whether entered number is prime or not with the help of user-defined function check-prime(). (Mar-2022)[NLJIET]	7	CO4	CR
6	Write a function in the C program to return 1 if number is prime otherwise return 0. (Aug-2023)[NLJIET]	7	CO4	CR
7	Write a program to calculate nCr using user defined function. $nCr = n! / (r! * (n-r)!)$. (Jun- 2019) [NLJIET]	7	CO4	CR
8	Write a function which accepts a string and returns the length of the string. [NLJIET]	7	CO4	CR
9	Write a function which takes 2 numbers as parameters and returns the gcd of the 2 numbers. Call the function in main (). [NLJIET]	7	CO4	CR
10	Develop a menu-based program to perform addition, multiplication, subtraction and division using user-defined function.(Mar-2022) [NLJIET]	7	CO4	CR
11	What is recursion? Write C program to find factorial (N) using recursive Function? [NLJIET]	7	CO4	CR
12	Write a C program to find factorial of a number using recursion (Aug-2023)[NLJIET]	7	CO4	CR
13	Apply merge sort algorithm for the following data and show the steps. [NLJIET] 66, 33, 40, 22, 55, 88, 11, 80, 20, 50, 44, 77, 30	7	CO4	CR
14	Write a C program to generate first n number of Fibonacci series using recursion (0 1 1 2 3 5 8 ...) (Aug-2023)[NLJIET]	7	CO4	CR
15	Write a recursive program to find factorial of a given number. (Aug-2022)[NLJIET]	7	CO4	CR
16	What is recursive function? Write a program to find out factorial of a number using recursive function.[NLJIET]	7	CO4	CR
17	Write a C Program to find sum of digits for a given number using the concept of User Defined Function UDF. (Jan -2024)[NLJIET]	7	CO4	CR
18	Construct a C program which used a user define function that receives an integer and returns square and cube of it and prints in main function.(Jul -2024)[NLJIET]	7	CO4	CR
19	Write a program in C to calculate the power of any number using recursion. (Jan -2025)[NLJIET]	7	CO4	CR

20	Develop a program in C to check the entered number is prime or not by creating a user-defined function named check_prime(). (Jan -2025-New)[NLJIET]	7	CO4	CR

CHAPTER: 5

ARRAYS AND STRINGS

Arrays: One-dimensional and multi-dimensional arrays, array operations. • Strings: String handling functions, basic string operations.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is array? Explain initialization of an array value by example. [NLJIET]	3	CO3	UN
2	What is array? Give example of array. [NLJIET]	3	CO3	AP
3	What is array? Give example and advantages of array. [NLJIET]	3	CO3	UN
4	Describe array with example. (Jan-2019) [NLJIET]	3	CO3	UN
5	Show 1D array declaration, initialization and iteration. (NOV-2020) [NLJIET]	3	CO3	AP
6	Show 2D array declaration, initialization and iteration. (NOV-2020) [NLJIET]	3	CO3	AP
7	Demonstrate declaration and initialization of two dimensional array with suitable example. (Mar-2022) [NLJIET]	3	CO3	AP
8	Why it is necessary to give the size of an array in array declaration? (Jun-2019) [NLJIET]	3	CO3	AN
9	Give the output of following C code.(Jan-2024)[NLJIET] <pre>int main() { int arr r[3][2] = {1, 2, 3, 4, 5, 6} ; printf(“%d %d %d”, arr[0][1], arr[1][0], arr[2][1]); return 0; }</pre>	3	CO3	UN
10	Explain how string is defined in C. List the various inbuilt string functions. [NLJIET]	3	CO3	RM
11	Give the significance of puts(), getchar(), getch(). (Jan-2020)(Jan-2024)[NLJIET]	3	CO3	UN
12	Explain strcat(),strlen() and strcpy() functions with examples.[NLJIET]	3	CO3	UN
13	Explain following string function with suitable example. [NLJIET] 1. strcmp 2. strcat	3	CO3	AP
14	Explain following string handling functions: I. strcat() II. strcmp()III.strlen (Aug-2023)[NLJIET]	3	CO3	UN

15	Distinguish between the following pairs: 1) getchar and scanf functions 2) %s and %c specifications for reading. 3) %s and %[] specifications for reading. [NLJIET]	3	CO3	AN
16	Distinguish between the following pairs: 1) getchar and scanf functions 2) %s and %c specifications for reading. [NLJIET]	3	CO3	AN
17	Give the significance of <i>getc()</i> , <i>getw()</i> , <i>fscanf()</i> . (Mar-2023)[NLJIET]	3	CO3	UN
18	Explain <i>getc()</i> and <i>putc()</i> functions with suitable example.[NLJIET]	3	CO3	AP
19	Write a program to reverse the input string. [NLJIET]	3	CO3	CR
20	Write a program to check whether entered character is vowel or not? (Mar-2022)(Jan-2025-New)[NLJIET]	3	CO3	CR
21	List the types of Array.(Jul -2024)[NLJIET]	3	CO3	RM
22	List the application of <i>gets</i> <i>getchar</i> and <i>puts</i> functions. (Jul -2024)[NLJIET]	3	CO3	RM
23	What is string ? How to declare string in C? (Jan-2025)[NLJIET]	3	CO3	UN
24	Explain significance of following string functions with example. 1. <i>strcat()</i> 2. <i>strlen()</i> 3. <i>strcpy()</i> 4. <i>strcmp()</i> (Jan-2025)[NLJIET]	4	CO3	AP
25	Summarize the methods for initialization of One-Dimensional array.(Jul -2024)[NLJIET]	4	CO3	UN
26	Summarize the method for Declaration, initialization and Printing on screen of a String.(Jul -2024)[NLJIET]	4	CO3	UN
27	Discuss initialization of one-dimensional arrays with example.[NLJIET]	4	CO3	EL
28	Why null value is used in string? Justify your answer with example. (Jan-2019)[NLJIET]	4	CO3	EL
29	What is string? In how many ways can you accept data in a string? [NLJIET]	4	CO3	UN
30	Explain any four inbuilt string functions. (Nov-2020) [NLJIET]	4	CO3	UN
31	What is string? Explain with example to find a character from string. (Jan-2019) [NLJIET]	4	CO3	AP
32	Explain <i>getch()</i> , <i>getchar()</i> , <i>gets()</i> , <i>puts()</i> . (Jan-2020)(Mar-2022)(Jan-2025-New)[NLJIET]	4	CO3	UN
33	Explain following string manipulation function. <i>strcat()</i> , <i>strcpy()</i> , <i>strcmp()</i> and <i>strlen()</i> (Mar-2022)(Jan-2025-New) [NLJIET]	4	CO3	UN
34	Explain following string functions :(1) <i>strcpy()</i> , (2) <i>strcat()</i> , (3) <i>strlen()</i> , (4) <i>strcmp()</i> [NLJIET]	4	CO3	UN
35	Explain <i>putchar()</i> and <i>puts()</i> functions with suitable example. [NLJIET]	4	CO3	AP
36	What is string? Explain <i>gets()</i> and <i>puts()</i> functions with example.[NLJIET]	4	CO3	UN

37	Write a program to check whether entered character is vowel or not? (Jun-2019) [NLJIET]	4	CO3	CR
38	Summarize the methods for initialization of One-Dimensional array. (Jan-2025-New) [NLJIET]	4	CO3	RM

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is an array? Explain one dimensional and two dimensional array declarations and initialization with suitable example. (Jun-2019) [NLJIET]	7	CO3	UN
2	What is String? How are they declared and also define the null character. [NLJIET]	7	CO3	UN
3	What is a string? Explain at least 4 built-in string functions with example. (Jun-2019) [NLJIET]	7	CO3	UN
4	Explain any four string handling functions with an example. (Mar-2023) [NLJIET]	7	CO3	UN
5	What is string? Write a program to concatenate two strings without using built in function. [NLJIET]	7	CO3	UN
6	What is string? Write a program to reverse the input string. [NLJIET]	7	CO3	CR
7	Describe following string function in C – Language. 1) strcat() 2) strcpy() 3) strcmp() 4) strlen() [NLJIET]	7	CO3	UN
8	List out all various string functions and describe them with syntax and example. (Aug-2022) [NLJIET]	7	CO3	UN
9	Explain different string functions with example. (Jun-2025-New) [NLJIET]	7	CO2	AP
10	Explain Declaration, initialization and Printing on screen of a String using suitable program. (Jun-2025-New) [NLJIET]	7	CO3	AP
11	Explain one dimensional array, two-dimensional array and multi-dimensional array with their syntax and example. (Jun-2025-New) [NLJIET]	7	CO3	AP

PROGRAMS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Write a program in c for multiplication of two matrix. [NLJIET]	4	CO3	CR
2	Write a program in c for multiplication of two matrix. [NLJIET]	7	CO3	CR
3	Write a C program to multiply two N X N Matrix. [NLJIET]	7	CO3	CR
4	Write a program in c for multiply two matrices A and B of dimensions pXq and qXr respectively and store the result in third matrix C. (Mar-2022) [NLJIET]	7	CO3	CR
5	Write a program to find out the largest of an array.	7	CO3	CR

	(Mar-2022) [NLJIET]			
6	Write a program to arrange the number of an array in ascending order. [NLJIET]	7	CO3	CR
7	Write a program to store 10 elements in array given by user and to find maximum out of those 10 elements. (Nov-2020) [NLJIET]	7	CO3	CR
8	Write a C program to find maximum and minimum from an array of 10 elements(Jan-2024)[NLJIET]	7	CO3	CR
9	Write a C program to read 10 numbers from user and store them in an array. Display Sum, Minimum and Average of the numbers. (Jan-2020) [NLJIET]	7	CO3	CR
10	Write a C program to get 20 numbers from user. Find sum, average, maximum and minimum number.[NLJIET]	7	CO3	CR
11	Write a program to display transpose of given 3*3 matrix. [NLJIET]	7	CO3	CR
12	Write a C program to sort an array in ascending order. (Mar-2023)[NLJIET]	7	CO3	CR
13	Write a C program to make sum of array elements. (Mar-2023)[NLJIET]	7	CO3	CR
14	Write a program to arrange N numbers of an array in ascending order [NLJIET]	7	CO3	CR
15	Write a program to accept a string and count the number of vowels present in a string.[NLJIET]	7	CO3	CR
16	Write a program to copy one string into another and count the number of characters copied. [NLJIET]	7	CO3	CR
17	Write a program to find a character from the string, string and character to be searched both will be given by user. (Nov-2020) [NLJIET]	7	CO3	CR
18	Write a function program to find whether the string is palindrome or not. [NLJIET]	7	CO3	CR
19	Write a program to count total word in text. [NLJIET]	7	CO3	CR
20	Write a C program to do following task. (1) Take string from user and print in upper-case (2) Find string length without using strlen() function [NLJIET]	7	CO3	CR
21	Write a C program to copy one string to other without using string handling function.(Aug-2023)[NLJIET]	7	CO3	CR
22	Write a C program to check whether two strings are same or not. (Jan-2024)[NLJIET]	7	CO3	CR
23	Construct a C program to multiply 3X3 matrix. (Jul-2024)[NLJIET]	7	CO3	CR
24	Write a C program ,that reads a statement from user and prints the frequency of each of the distinct character from a-z. (Jan-2025)[NLJIET]	7	CO3	CR

25	Write a C program which takes two NxN matrices and performs matrix multiplication. (Jan-2025)[NLJIET]	7	CO3	CR
26	Write a program to add all the elements of an array. (Jan-2025-New)[NLJIET]	7	CO3	CR
27	Construct a C program to add 3X3 matrix. (Jan-2025-New)[NLJIET]	7	CO3	CR

CHAPTER: 6 POINTERS

Pointers: Basics of pointers, pointer arithmetic, pointers and arrays, pointers to functions (for C)

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is a pointer? How is a pointer initialized? [NLJIET]	3	CO3	UN
2	What is a pointer? Explain initialization of a pointer in details [NLJIET]	3	CO3	UN
3	What is Pointer in C? Explain its advantages. [NLJIET]	3	CO3	UN
4	What is pointer? Explain how pointers are declared and initialized. (Mar-2022- NEW) [NLJIET]	3	CO3	UN
5	What is a pointer? How and when is it used? [NLJIET]	3	CO3	UN
6	What is pointer? Explain with example to print the address of variable using pointer. (Jan-2019) [NLJIET]	3	CO3	UN
7	What is pointer? Explain with example to store and print the address of variable using pointer. (Nov-2020) [NLJIET]	3	CO3	UN
8	What is pointer? Which arithmetic operations are not valid on pointers? (Jun-2019) [NLJIET]	3	CO3	UN
9	What is pointer? Which arithmetic operations are valid on pointers? (Jun-2019)[NLJIET]	3	CO3	UN
10	What is meant by array of pointers? Explain it with example.(Aug-2022)[NLJIET]	3	CO3	UN
11	What is pointer? Explain how pointers are declared and initialized. State its advantages.[NLJIET]	3	CO3	UN
12	Write True / False against following statements: 1) Pointer is a variable which holds address of location where value of other variable is stored. 2) Pointer holds the address in form of float or integer. 3) The address of a variable can be stored in two or more pointers.(Jul -2024)(Jan-2025-New)[NLJIET]	3	CO3	AN

13	What are the arithmetic operators that are permitted on pointers?(Jul -2024)[NLJIET]	3	CO3	UN
14	What is a pointer? What are the advantages of using pointer? (Jan-2025)[NLJIET]	3	CO3	UN
15	What is pointer? Which arithmetic operations are valid on pointers? (Jun-2025-New) [NLJIET]	3	CO4	UN
16	What is pointer to pointer? Write suitable example to demonstrate the concept. (Jun-2025-New) [NLJIET]	3	CO4	UN
17	What is pointer? Explain how pointers are declared and initialized. State its advantages.[NLJIET]	4	CO3	UN
18	What is pointer to pointer? Write suitable example to demonstrate the concept.(Jun-2019) [NLJIET]	4	CO3	AP
19	What is Pointer in C and how it initializes? State its advantages. (Aug-2023)[NLJIET]	4	CO3	UN
20	How to initialize Pointer? Explain arrays of pointer using suitable sample program. [NLJIET]	4	CO3	AP
21	Explain array of pointers with suitable example. (Jun-2019) [NLJIET]	4	CO3	AP
22	What is pointer? Explain Array of pointer with example. [NLJIET]	4	CO3	AP
23	Explain one dimensional array with example [NLJIET]	4	CO3	UN
24	What is pointer variable? Write syntax to create a pointer variable that points to an array of 20 float numbers. [NLJIET]	4	CO3	CR

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is Pointer? How is Pointer initialized? How is it different from Array? [NLJIET]	7	CO3	UN
2	How to initialize integer pointer? Describe pointer to array with example. [NLJIET]	7	CO3	UN
3	What is pointer? Give its benefits. Write a program to do swapping of two elements using function with two pointers as arguments. [NLJIET]	7	CO3	CR
4	Explain one dimensional array, two dimensional array and multi-dimensional array with their syntax and example. (Aug-2022)[NLJIET]	7	CO3	AP

PROGRAMS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Write a program to do swapping of two elements using function with two pointers as arguments. [NLJIET]	4	CO3	CR

2	Write a program using pointer and function to determine the length of string. [NLJIET]	4	CO3	CR
3	Write a program to read n different integer numbers from keyboard and calculate the sum using pointer. [NLJIET]	7	CO3	CR
4	Write a C program to find the sum and mean of all elements in an array using pointer. (Jan-2025)[NLJIET]	7	CO3	CR

CHAPTER: 7 STRUCTURES AND UNIONS

Structures: Defining and using structures, array of structures, pointer to structures. • Unions: Basics and usage.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is a structure? Give example. [NLJIET]	3	CO3	UN
2	What is structure? Explain with example how to declare a structure and how to initialize it. (Mar-2022)(Jan-2025-New) [NLJIET]	3	CO3	UN
3	Distinguish between Structure and Union. (Jan-2020) [NLJIET]	3	CO3	UN
4	State difference between Union and Structure. [NLJIET]	3	CO3	UN
5	Differentiate structure and union with suitable example. [NLJIET]	3	CO3	UN
6	Differentiate between structure and union.(Jan-2024)(Jan-2025-New) [NLJIET]	3	CO3	UN
7	Give merits and demerits of Union over structure. [NLJIET]	3	CO3	UN
8	Define Union in 'C' with example. (Aug-2022)[NLJIET]	3	CO3	UN
9	Compare Array, Structure and Union. (Jan-2025)[NLJIET]	3	CO3	UN
10	Define Union. Describe how to declare, initialize and access members of Union with a programming example. (Jan-2025)[NLJIET]	4	CO3	UN
11	Show structure declaration and initialization with proper syntax. (Nov-2020)[NLJIET]	4	CO3	AP
12	Explain how structure variable is initialized with suitable example. (Jun-2019)[NLJIET]	4	CO3	UN
13	Illustrate the concept of "structure"? Explain how can we access structure members. [NLJIET]	4	CO3	AP
14	Distinguish between "structure" and "array". Discuss the meaning and purpose of following: (I) struct keyword (II) typedef keyword (III) sizeof operator. [NLJIET]	4	CO3	UN

15	Design structure with following elements: Name of employ, basic, da, ta, deduction, net_salary Give the variety of way to access individual elements of structure. [NLJIET]	4	CO3	CR
16	Explain with suitable example structure variable and pointer to structure variable.(Jun- 2019) [NLJIET]	4	CO3	UN
17	Write the difference between structure and union. [NLJIET]	4	CO3	UN
18	Compare structure and union.(Mar-2023)[NLJIET]	4	CO3	UN
19	What is structure? How it is different from union? Explain nested structure with example.(Aug-2023)[NLJIET]	4	CO3	UN
20	Explain structure within structure with example. (Aug-2022)[NLJIET]	4	CO3	UN
21	What is structure? How it is different from union? Explain nested structure with example. (Jun-2025-New) [NLJIET]	4	CO4	UN
22	Define Union in 'C' with suitable example. (Jun-2025-New) [NLJIET]	4	CO4	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is structure? Explain declaration and initialization of structure values with example. [NLJIET]	7	CO3	UN
2	What is structure? How to access the elements of structure? How to calculate size of structure? Explain with example. (Jan-2019) [NLJIET]	7	CO3	UN
3	Which type of problem can be solved by structure? Explain it with C program. [NLJIET]	7	CO3	UN
4	What is structure? Explain nested structure and array of structure with example. [NLJIET]	7	CO3	UN
5	Explain nested structure and array of structure with example. [NLJIET]	7	CO3	UN
6	What is structure? How does a structure differ from an array? How is an array of structure initialized? [NLJIET]	7	CO3	UN

PROGRAMS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Define a structure called cricket that will describe the following information: Player name Team name Batting average [NLJIET]	7	CO3	CR
2	Define a structure data type called time_struct containing three member's integer hours, minutes, second. Develop a program that would assign values to individual member and display the time in following format : HH:MM:SS (Jan-2019) [NLJIET]	7	CO3	CR

3	Define a structure person, that would contain person name, date of joining and salary. Using this structure, write a program to read this information for one person from the key board and print the same on the screen. [NLJIET]	7	CO3	CR
4	Write a program in c using structure to enter rollno, marks of the three subject for 3 student and find total obtained by each student. (Mar-2022) [NLJIET]	7	CO3	CR
5	Define a structure “personal” that would contain person name, date of joining and salary. Using this structure read information of 5 people and print the same on screen. Also display sum of salary of all 5 people. (Nov-2020) [NLJIET]	7	CO3	CR
6	Define a structure type struct personal that would contain person name, date of joining and salary using this structure to read this information of 5 people and print the same on screen. (Aug-2023) [NLJIET]	7	CO3	CR
7	Define a Structures which contains details of a Cricketer: • Name of Player: • Team name: • Total run scored: • Batting average: (Jul -2024)(Jan-2025-New) [NLJIET]	7	CO3	CR

CHAPTER: 8 FILE HANDLING

File Operations: Opening, closing, reading, writing files. • File Pointers: File pointers and basic file operations (for C).

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Describe file management? Recall various file modes. [NLJIET]	3	CO3	UN
2	What is a file? List the various file handling operations in c. [NLJIET]	3	CO3	RM
3	Describe following file handling functions. (1) fopen() (2) fseek() (3) fgetc(). [NLJIET]	3	CO3	UN
4	Explain various file management function in C. (Nov-2020) [NLJIET]	3	CO3	UN
5	Distinguish between the following functions: (I) “getc” and “getchar” (II) “printf” and “fprintf” (III) “feof” and “ferror”. [NLJIET]	3	CO3	UN
6	What is File Pointer? What is significance of File Pointer? (Jan-2025) [NLJIET]	3	CO3	RM
7	Explain various file management function in C. (Jun-2025-New) [NLJIET]	3	CO3	UN
8	Explain signification of following functions in file operations.	4	CO3	UN

	1. fseek() 2. ftell() 3. open() 4. fread() (Jan-2025) [NLJIET]			
9	Describe file management? List the various file handling operations in c. [NLJIET]	4	CO3	UN
10	Describe file management. And List the various file management functions. (Jan-2020) [NLJIET]	4	CO3	UN
11	Write syntax of fseek() function and explain fseek(fp,-10,1) and fseek(fp,10,0). [NLJIET]	4	CO3	CR
12	Explain the use of fputc() and fputs() with suitable example. [NLJIET]	4	CO3	UN
13	Explain fopen() and its mode with example to write a string into file. (Jan-2019)(Mar-2022)[NLJIET]	4	CO3	UN
14	Briefly explain any two file handling functions with an example (Jan-2024)[NLJIET]	4	CO3	UN
15	Explain getw() and putw() function with example. [NLJIET]	4	CO3	UN
16	Explain fopen() and its mode with example. (Jan-2025-New) [NLJIET]	4	CO3	UN

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is file management? List the different file management functions and explain the various file modes. [NLJIET]	7	CO3	RM
2	What is File Management? Explain various file management functions? [NLJIET]	7	CO3	UN
3	Explain the following File Handling functions: 1. fseek() 2. ftell() 3. fread() 4. fwrite() 5. fscanf() 6. fprintf() 7. rewind().(Aug-2022)[NLJIET]	7	CO3	UN
4	Explain error handling in file system with example. (Aug-2022)[NLJIET]	7	CO3	UN
5	List and discuss file handling functions used in C. (Jul -2024)[NLJIET]	7	CO3	RM
6	Briefly explain any two file handling functions with an example. (Jan-2025-New) [NLJIET]	7	CO3	UN
7	Explain error handling in file system with example. (Jun-2025-New) [NLJIET]	7	CO3	AP
8	Explain the following File Handling functions: 1. fseek() 2. ftell() 3. fread() 4. fwrite() 5. fscanf() 6. fprintf() 7. rewind(). (Jun-2025-New) [NLJIET]	7	CO3	AP

PROGRAMS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Write a program to illustrate the use of fputc () and fputs().	3	CO4	CR

	(Jan-2020) [NLJIET]			
2	Write C program to copy content of file “D:\home\avs.txt” into new text file “ftel.txt” in E drive. [NLJIET]	4	CO4	CR
3	Write a program to illustrate the use of fputc () and fputs(). [NLJIET]	4	CO4	CR
4	Write a C program to copy content of one file to other with the help of file handling functions.(Mar-2023)(Jan-2025-New)[NLJIET]	4	CO4	CR
5	Write a ‘C’ program to which copies the contents of one file to other. [NLJIET]	7	CO4	CR
6	Write C program to copy one file to other.(Jan-2024)[NLJIET]	7	CO4	CR
7	Write a program which reads content of a file named “content.txt” and find out the number of vowels in the file.[NLJIET]	7	CO4	CR
8	Write a C program to read ID and marks of n number of students from user and store them in a file. (Jan-2025)[NLJIET]	7	CO4	CR

CHAPTER: 9**DYNAMIC MEMORY ALLOCATION**

Memory Allocation: malloc, calloc, realloc, and free functions. Usage and examples.

SHORT QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	Explain dynamic memory allocation. [NLJIET]	3	CO3 CO4	UN
2	What is dynamic memory allocation? Show the use of malloc() and calloc() function with their syntax (Aug-2023)[NLJIET]	3	CO3 CO4	UN
3	Explain the use if malloc and calloc function with their syntax. [NLJIET]	3	CO3 CO4	UN
4	Describe dynamic memory allocation. Give advantages and disadvantages of it. [NLJIET]	3	CO3 CO4	RM
5	List down and briefly explain methods for dynamic memory allocation.(Mar-2023)[NLJIET]	3	CO3 CO4	RM
6	Explain dynamic memory allocation [NLJIET]	3	CO3 CO4	UN
7	Explain need of Dynamic Memory allocation. Compare malloc(), calloc() and realloc() functions. (Jan-2025)[NLJIET]	3	CO3 CO4	UN

8	Compare malloc() and calloc() functions for dynamic memory allocation. (Mar-2022) [NLJIET]	4	CO3 CO4	UN
9	Recall the definition of dynamic memory allocation? Differentiate between malloc and calloc? [NLJIET]	4	CO3 CO4	RM
10	What is dynamic memory allocation? Show the use of malloc () and calloc () function with their syntax. [NLJIET]	4	CO3 CO4	AP
11	Explain malloc (), calloc () with examples. [NLJIET]	4	CO3 CO4	UN
12	Show the usage of malloc () and calloc () with syntax. (Nov-2020) [NLJIET]	4	CO3 CO4	AP
13	Define dynamic memory allocation. Explain malloc() and calloc().(Aug-2022)[NLJIET]	4	CO3 CO4	RM
14	Briefly explain memory management functions. (Jan-2024)[NLJIET]	4	CO3 CO4	UN
15	Show the use of Malloc, Calloc, Free and Realloc in context of Dynamic memory allocation.(Jul -2024)[NLJIET]	4	CO3 CO4	AP
16	Show the use of Malloc, Calloc, Free and Realloc in context of Dynamic memory allocation. (Jun-2025-New) [NLJIET]	4	CO3 CO4	AP

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is dynamic memory allocation? Explain important functions associated with it.(Jun-2019) [NLJIET]	7	CO3 CO4	RM
2	Explain dynamic memory allocation with example. [NLJIET]	7	CO3 CO4	UN

CHAPTER: 10**DEBUGGING AND TESTING**

Debugging Techniques: Common debugging practices, use of debugging tools. • Testing: Writing test cases, unit testing

SHORT QUESTIONS

Sr.	QUESTIONS	Marks	CO	RBT
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No.				Level
1	Explain white box testing.[NLJIET]	3	CO5	UN
2	Elaborate in detail- black box testing.[NLJIET]	3	CO5	UN
3	Explain debugging techniques. (Jun-2025-New) [NLJIET]	3	CO5	UN
4	Explain types of errors in detail.[NLJIET]	4	CO5	UN
5	What is debugging? Explain techniques of debugging in detail.[NLJIET]	4	CO5	UN
6	Define following: 1) IDE Debugging 2)Command line Debugging[NLJIET]	4	CO5	RM

DESCRIPTIVE QUESTIONS

Sr. No.	QUESTIONS	Marks	CO	RBT Level
1	What is testing? Explain Types of testing in detail.[NLJIET]	7	CO5	UN

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